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Investigating the effect of different fine-tuning configuration scenarios on agricultural term extraction using BERT

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ABSTRACT

This paper compares different transformer-based language models for automatic term extraction from agriculture-related texts. Agriculture is an important economic sector faced with severe environmental and societal challenges. The collection, annotation and sharing of agricultural scientific knowledge is key to enabling the agricultural sector to address its challenges. Automatic term extraction is a Natural Language Processing task that can provide solutions to text tagging and annotation towards better knowledge and information exchange. It is concerned with the identification of terms pertaining to a domain, or area of expertise, in text and is an important step in knowledge base creation and update pipelines. Transformer-based language modeling technologies like BERT have become popular for automatic term extraction, but limited work has been undertaken so far in applying these methods to agriculture. This paper systematically compares Agriculture-BERT to Sci-BERT, RoBERTa, and vanilla BERT, which were fine-tuned for the automatic extraction of agricultural terms from English texts. The greatest challenge faced in our research was the scarcity of agriculture-related gold standard corpora for measuring automatic term extraction performance. Our results show that, with a few exceptions, Agriculture-BERT performs better than the other models considered in our research. Our main contribution and novelty of the presented research is the investigation of the impact that different language model fine-tuning configuration scenarios had on the term extraction task. More specifically, we tested different scenarios related to the model layers kept frozen, or being updated, during training, to measure the impact they may have on Agriculture-BERT's performance in automatic term extraction. Our results show that the best performance was achieved by: (i) the "embedding layer updated + all encoder layers updated" scenario for the identification of terms also seen during training; (ii) the "embedding layer frozen + all encoder layers updated" scenario for the identification of terms being synonyms to those seen during training; and (iii) the "embedding layer updated + top 4 encoder layers updated" scenario for identifying terms neither seen during training nor being synonyms to those seen during training (novel terms).

1. Introduction

Automatic term extraction (ATE), also known as automatic term recognition, terminology mining, or glossary extraction, concerns the identification of single- or multi-word terms, associated with a domain or topic, in unstructured text (Zhang et al., 2008). It is a task where each word is tagged with a label indicating whether the word is (part of) a domain term or not and differs from Named Entity Recognition (NER) in which words are tagged with labels denoting specific entity types. ATE is an important step in Natural Language Processing (NLP) tasks including automatic knowledge base construction/update, semantic search, document classification and indexing, text summarization, question answering, and topic detection (Pazienza et al., 2005; Astrakhantsev et al., 2015). In this paper we focus on the automatic extraction of agriculture-related terms from English texts using transformer-based language models fine-tuned on the specific task.

Despite a large body of existing research, most term extraction methods, used in various domains, have variable performance with

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considerable room for improvement (Rigouts Terryn et al., 2018). Part of the issue is that determining whether a string forms a domain-relevant term is frequently very subjective (Pazienza et al., 2005; Rigouts Terryn et al., 2018). This vagueness is rooted in the fact that there is no consensus on what constitutes a term related to a domain (Pazienza et al., 2005; Rigouts Terryn et al., 2018). Operational definitions found in literature describe "term" as a lexical representation of a concept, i.e., an abstract mental construct, relevant to a domain (Wright, 1997). However, this definition is very broad leaving room for different interpretations. Given the cross-disciplinary nature of almost any domain and area of expertise, it is difficult to use this definition for deciding about the "termhood" of a text span.

Early efforts in ATE were based on methods using manually crafted features or rules to compute the pertinence of a word or phrase to a domain, as well as unsupervised methods generating clusters of semantically similar words found in text (Astrakhantsev et al., 2015; Tran et al., 2023). Recent advances in deep neural network technologies provide affordances for learning the context in which terms occur and thus better performing ATE methods. Recently, transformers have become popular for the task of ATE (Tran et al., 2023). They enable language model pre-training using specialized text corpora, which has proven to lead to increased performance in various NLP tasks (Dai and Le, 2015). As opposed to unidirectional language models where a token only attends to previous tokens in the self-attention layer, transformer-based models (e.g., BERT) enable the bidirectional capture of the context in which a token occurs (Devlin et al., 2018). This capacity leads to better performance in NLP tasks when fine-tuning a pre-trained language model on that specific task (e.g., ATE or NER).

Agriculture is an economic sector of major significance accounting for 4 % of the global Gross Domestic Product (World Bank, 2022) as well as providing income to nearly 20 million people working on the 10.3 million farms in Europe (Eurostat, 2022). Agriculture needs to shift to more sustainable and climate friendly practices to reduce its environmental footprint which is responsible for 25 % of greenhouse gas emissions at a global scale and devastating impacts on biodiversity (World Bank, 2022). Digital technologies have the potential to provide solutions for more environmentally friendly and sustainable agrifood production (Kamilaris et al., 2017; Sharma et al., 2022). The penetration of digital technologies in agriculture in the past few years has turned agrifood production into a knowledge-intensive process with the demand for standardized data and information exchange continuously increasing (Ngo et al., 2022).

Agriculture-related ontologies and knowledge graphs provide standardized means for intra- and cross-domain knowledge exchange and interoperability (Jonquet et al., 2018). Therefore, keeping them updated by timely capturing developments is a critical enabler for the advancement of sustainable agricultural practices. Given the pace and volume at which new information and data become available from agrifood-related publications, the use of information extraction methods provides a valuable support to keeping track of advancements and integrating them to domain knowledge representations (Drury and Roche, 2019). ATE allows for the identification and extraction of (novel) terms from text and is thus important for knowledge base update. However, NLP research in agriculture is still in its infancy despite the rapid steps taken in the digitalization of the agricultural sector.

Considering the increased interest in transformer-based language models for ATE and their advantages compared to other rule-based and statistical methods (Tran et al., 2023), as well as BERT's capacity for bidirectional context capturing, our research focuses particularly on BERT for the task of agricultural term extraction from English texts. Traditional term extraction methods use hand-crafted generic rules for the extraction of terms, not necessarily pertaining to patterns of agriculture-related term occurrence, by being provided with domain-specific texts as input. These rules normally draw on statistical features quantifying importance (e.g., TF-IDF) and/or features that capture syntactic structures associated with term occurrence in general. Transformer technologies like BERT provide affordances for domain term identification from specialized texts by learning the context in which terms occur without involving any hand-crafted feature engineering at all. This capacity makes transformer technologies competitive when used for term extraction purposes and better performing when compared to non-neural term extractors (Tran et al., 2023). For these reasons, we decided to explicitly focus on transformers for the creation of our agriculture-related term extractor and not consider traditional, non-neural term extraction methods. Our research questions are:

- Does a BERT language model pre-trained on agricultural texts perform better than similar or general-purpose BERT models when fine-tuned for automatic agriculture-related term extraction?
- What is the effect of different fine-tuning configuration scenarios, related to model layers updated during training, on ATE performance?

Our research systematically compares the performance of different BERT models in the extraction of agriculture-related terms from English texts. We compare Agriculture-BERT (BERT model pre-trained on agricultural texts) (Quadros, 2021), Sci-BERT (Beltagy et al., 2019), RoBERTa (Liu et al., 2019), and vanilla BERT (Devlin et al., 2018). The last two models were used as baselines. The challenge we had to face was the lack of gold standard corpora to measure performance in agricultural term extraction. However, the scarcity of gold standard corpora is not limited to the domain of agriculture. It is a problem related to benchmarking ATE methods in general, regardless of the domain of application (Tran et al., 2023). In addition, the creation of benchmark datasets by human annotators is challenging because of the increased degree of subjectivity that is inherent to determining whether a word or phrase makes an agricultural term, or not. For these reasons, a silver standard corpus consisting of publication abstracts indexed in the AGRIS database[1] was used to train and evaluate all the models compared in our research. The AGRIS abstract texts were automatically tagged for the presence of agriculture-related terms in them using the concepts defined in AGROVOC[2] as a silver standard of terms pertaining to agriculture. Silver standard corpora are built using existing vocabularies and automatic text tagging methods, and constitute an alternative to gold standard corpora, given that gold standard corpora creation is expensive and/or may suffer from low annotator agreement (Rashed et al., 2020).

We measured the performance of the BERT models in the identification of unseen terms (i.e., terms not seen during training) to determine their generalization capacity. The ability of a system to identify terms not seen during training is important if we consider that a term may have different morphological forms, as well as that novel terms constantly emerge. For instance, "fattening pigs" is a term introduced to AGROVOC as a new concept in 2019, yet there had already been almost 1.7 K occurrences of it in AGRIS publications dating before 2019 (Panoutsopoulos and Brewster, 2022). To this end, our silver standard corpus was split to a training set, a validation set, and three test sets (memorization, synonym, and novel term test sets). It consisted of nearly 17 K sentences (training set: 11.5 K sentences; validation set: 1.5 K sentences; test sets: ~ 4 K sentences). One of the test sets (the memorization term set) was used to measure performance in the extraction of terms also seen during training (memorization capacity). The other two test sets (the synonym and novel term sets) were used to measure performance in unseen term identification (generalization capacity). Our results show that with a few exceptions Agriculture-BERT had the best performance measured on all our test datasets (memorization, synonym, and novel term sets).

The main contribution of our research is the investigation of the impact of different language model fine-tuning configuration scenarios

[1] https://www.fao.org/agris/.
[2] https://www.fao.org/agrovoc/.

on the capacity of Agriculture-BERT to identify agricultural terms in text. For this purpose, we tested different configuration scenarios regarding the model layers kept frozen, or being updated, during training and measured term extraction performance. We found that different scenarios had the best performance in each of the model test cases considered in our research, which related to identifying terms seen during training and terms not seen during training (i.e., synonym and novel terms).

The remainder of the paper is structured as follows: Section 2 provides an overview of research studies on ATE in agriculture. In section 3, we describe the methods used in our research with a focus on text preprocessing and tagging for building our silver standard corpus. The architecture and fine-tuning of our ATE are also explained. Our datasets, experimental setup, and results of the conducted experiments are described in Section 4. In Section 5, we discuss the results of our experiments. Conclusions focus on the strengths and limitations of our research, as well as on future research directions. The code and datasets created are available in the paper's GitHub repository.[3]

2. Related work

Research in agriculture-related ATE is sparse, mostly focusing on the investigation of how well domain-agnostic term extraction algorithms and tools can perform when used for the extraction of agricultural terms from text. Despite the wide use of deep neural networks, and especially transformers, for ATE in domains like Biomedicine, Mathematics, Energy, Biomechanics, Chemistry, and Linguistics (Tran et al., 2023), the adoption of such approaches in the field of agriculture is still in its infancy. There are, however, a few studies focusing on methods using hand-crafted rules. Kaushik and Chatterjee (2018) describe the extraction of agriculture-related terms from Hindi texts using RENT. RENT (Chatterjee and Kaushik, 2017) is a tool that uses patterns in the form of regular expressions for the extraction of single- and multi-word terms. To evaluate term extraction performance, texts from Indian government websites were used. Another tool for ATE from agricultural texts is AgroTagger (Celli et al., 2015). It is a tool assigning AGROVOC labels to strings in text, which was trained on a corpus of 780 agriculture-related texts manually tagged with AGROVOC concept labels. It was tested on a set of 32 documents randomly selected from AGRIS, already assigned AGROVOC concept labels as semantic tags.

Chatterjee and Kaushik (2020) compare four term extraction tools: RAKE, TermRaider, TerMine, and RENT. RAKE (i) identifies candidate keywords/keyphrases; (ii) computes importance scores for the candidate keywords/keyphrases; and (iii) ranks keywords/keyphrases based on their importance scores (Rose et al., 2010). TermRaider is a domain-agnostic term extraction tool using linguistic and statistical methods based on TF-IDF scores (Cunningham et al., 2002). TerMine[4] makes use of the C-value/NC-value method (Frantzi et al., 2000) focusing on multi-word term identification. The evaluation of the compared tools was made on a corpus of texts taken from the Indian department of agriculture. The best performance was achieved by RENT.

Although research in agriculture-related ATE is dominated by rule-based methods, there have been also efforts using deep neural networks. A method based on BERT (Encoder-Attention-Based Automatic Term Recognition) is proposed by Manjunath and McCrae (2021). In that study, manually annotated Wikipedia articles were used for training. Approximately 800 agricultural documents from FAO were used for testing. Panoutsopoulos et al. (2022) propose ATE from agricultural texts using Python's spaCy library (Honnibal and Montani, 2017). The approach is based on the use of the language models shipped by default with spaCy together with the spaCy library's tokenizer and NER components. The NER component was re-trained on agricultural texts. A set of nearly 1,000 randomly selected abstracts from the AGRIS database, annotated by a group of experts, were used for training, validation, and testing. The study reports a relatively low inter-annotator agreement rate (41 %).

Espejo-Garcia et al. (2019) propose a deep neural network architecture for the extraction of crop treatment regulation related information from text (spatiotemporal and frequency-related information for pesticide application) to use it for the generation of machine-interpretable rules for Farm Management Information Systems (FMISs). The architecture proposed consists of three layers: (i) a sequence representation layer (using character or word embeddings); (ii) a layer modeling the context in which words occur (using bidirectional LSTMs or CNNs); and (iii) an inference layer (using a CRF or a Softmax activation function). A total number of eight different architectures are compared with the best performing one (character embeddings + bidirectional LSTMs + Softmax) achieving an F1 score of 88.3 %.

Rezayi et al. (2022) present AgriBERT, a BERT-based model pre-trained on texts from agrifood-related scientific publications to perform the task of semantically matching food descriptions to nutrition data. The model's fine-tuning on the semantic matching task was made using the FoodOn ontology (Dooley et al., 2018) as an external knowledge source. A method combining the use of BERT for the identification of the semantic context in which words occur and CNNs for the capture of morphological, glyph features of character symbols is proposed by Guo et al. (2022) for the task of agricultural NER in Chinese texts. Combining dynamic word embeddings with visual elements of the characters of the Chinese language, the authors provide evidence for increased NER performance (including capabilities for rare agricultural entity recognition) by reporting F1 scores exceeding 95 %.

Table 1 summarizes the performance reported in the studies reviewed in our research. As made evident from that table, there is a wide span in the performance of the reviewed methods. Regarding rule-based methods, there is a span greater than 45 % in precision (from 32.93 % to 79.92 %) and almost 45 % in recall (from 21.51 % to 65.27 %). In the case of (deep) neural network methods, precision ranges from 47.82 % to 63.35 % (a span of 15 %). Differences in recall are higher (more than 30 %), which lead to similar differences in F1 scores.

Table 1
Performance scores reported in the ATE studies reviewed in our research.

Tool/algorithm	Author(s)	Automatic term extraction performance		
		Precision	Recall	F1-score
RENT	Kaushik and Chatterjee (2018)	75.7 %	65.27 %	–
Extraction of crop regulation information from text	Espejo-Garcia et al. (2019)	–	–	88.3 %
RENT	Chatterjee & Kaushik (2020)	79.92 %	45.11 %	–
RAKE		52.49 %	40.72 %	
TerMine		54.35 %	21.51 %	
TermRaider		41.08 %	29.35 %	
Encoder-Attention-Based Automatic Term Recognition	Manjunath and McCrae (2021)	63.35 %	16.34 %	14.88 %
Dynamic word embeddings with glyph feature capture	Guo et al. (2022)	–	–	> 95 %
spaCy-based ATE	Panoutsopoulos et al. (2022)	47.82 %	49.22 %	48.45 %

(*) Author(s) of the research study in which the ATE tool/algorithm was evaluated.

[3] https://github.com/herculespan/BERT-based_ATE_for_agriculture.
[4] https://www.nactem.ac.uk/software/termine/.

In case of development of rule-based term extraction methods from scratch or fine-tuning of existing, general-purpose methods for domain-specific terminology extraction, the cost in human labor for rule definition increases. Normally, there is an increased cost in terms of the resources needed for the manual identification of the syntactic structures and linguistic patterns related to term occurrence (Tran et al., 2023). In addition, not all patterns can be captured and modelled. As opposed to rule-based methods, deep neural networks can learn the context in which terms occur in text and be used for well-performing ATE. However, not much research on agriculture related ATE using deep learning has been undertaken till now.

The availability of pre-annotated text corpora is important for the evaluation of the performance of term extraction methods. In almost all studies reviewed in our research, gold standard corpora were used for measuring ATE performance. However, there is little evidence of how these corpora were built. This is an issue related to the lack of benchmarks for ATE evaluation, hence resulting in the creation of custom datasets fitting the scope of each study. Given the cross-disciplinary nature of the agricultural domain, there is lots of subjectivity, and hence vagueness, in determining whether a string is a term related to agriculture (Panoutsopoulos et al., 2022). Consequently, the use of gold standard corpora for evaluation purposes may prove to be problematic unless a rigorous corpus creation process is followed. As discussed below, silver standard corpora appear to be a viable, alternative option.

3. Materials and methods

3.1. Retrieval of agriculture-related concept labels for text tagging

In our research, the labels of concepts defined in AGROVOC were used as a silver standard of terms related to agriculture for tagging the texts in our training, validation, and test datasets. AGROVOC is a widely known and broadly adopted vocabulary of more than 40 K agrifood-related concepts, containing definitions of concepts and concept labels in a wide range of languages (Subirats-Coll et al., 2022). It is structured as a directed acyclic graph based on the SKOS specification (Miles and Bechhofer, 2009) and represents associations between concepts in the form of hierarchical and non-hierarchical relations. Each concept in AGROVOC is an instance of the skos:Concept class uniquely identified by a URI and is named using preferred and alternative labels. More specifically, each concept has one preferred label and zero or more alternative labels in English. For instance, http://aims.fao.org/aos/agrovoc/c_5871 is the URI of the concept that has "piggeries" as its preferred label in English. Alternative labels of the concept (in English) are "pig housing", "swine housing", and "swine stables".

Preferred labels of AGROVOC concepts were used to tag the texts in our training and validation sets, as well as the memorization term set (i.e., one of our test sets described in the section "Experiments and results"). To obtain the preferred labels of AGROVOC concepts, we created a SPARQL query defining a pattern for the retrieval of all instances of the skos:Concept class with a preferred label in English (i.e., literals being values of the skos:prefLabel property and tagged as "@english") and ran it to AGROVOC's SPARQL endpoint.[5] After being pre-processed, the retrieved labels were used to tag the texts. Pre-processing concerned the removal of the parentheses appearing in some labels. Parentheses include: (i) acronyms (e.g., North Central States (USA)) or acronym expansions/explanations (e.g., TRAFFIC (wildlife trade monitoring network)); or (ii) additional information about the preferred label (e.g., Ariidae (14102XXXXX) or function testing (diagnosis)). In the first case, the string contained in parentheses was kept as a separate preferred label, whereas, in the latter case, it was discarded. Denoting the set of the preferred labels of the AGROVOC concepts used for the tagging of the texts in our training, validation, and memorization term sets as $\mathscr{L}_{pref}$, we have $|\mathscr{L}_{pref}| = 3{,}595$. The full list of the preferred labels of AGROVOC concepts that were used for text tagging is available in the paper's GitHub repository.[6] The preferred labels used for tagging the texts in each of the training, validation, and memorization term sets were subsets of $\mathscr{L}_{pref}$, such that the intersection of these three sets by two was not the empty set.

According to Kim and Kang (2022), machine learning models trained on the extraction of terms related to a domain may not perform well when used to identify terms not seen during training. This can be attributed to the fact that terms have synonyms or morphological variants, as well as that novel terms may occur in literature. To this end, the text in two of our test sets (the synonym and novel term sets) was tagged for the occurrence of terms considered synonyms of those seen during training, as well as novel terms (terms neither seen during training nor being synonyms of the terms seen during training).

There are multiple semantic relations that hold between words, which can be captured and defined in a formal way in knowledge representation structures. Apart from synonymy, such relations are antonymy, paronymy, metonymy, meronymy, and hypernymy/hyponymy. Terms are words or phrases with a specialized meaning, denoting concepts within a domain of application or area of expertise. In that case, relations like paronymy, antonymy, and metonymy are not considered relevant. The semantics built in ontologies and vocabularies allow for capturing synonymy and meronymy. As an example, we can mention the skos:related and skos:member properties which are used in AGROVOC for capturing synonymy and meronymy relations. However, a focus on terms associated with semantic relations other than synonymy is relevant to term and between term semantic relation extraction for knowledge base construction/update. It goes beyond the scope of our research in which performance in the identification of terms not seen during training had to be evaluated. By drawing on existing research, we decided to measure the generalization capacity of the BERT models by gradually shifting to terms being synonyms to those seen during training, as well as terms not having a semantic relation to terms seen in training. The assumption is that synonym terms will appear in similar textual contexts, whereas novel terms will most probably not appear in a similar context. As a result, there is potential to evaluate the competence of the models in learning features related to the context in which agricultural terms occur and identifying them in text.

The synonym and novel terms used to tag the texts in the respective test sets were alternative labels of AGROVOC concepts. Specifically, $\mathscr{L}_{syn}$ is the set of synonym terms (alternative labels of the concepts in AGROVOC whose preferred labels were contained in $\mathscr{L}_{pref}$) used to tag the texts in the synonym term set. $\mathscr{L}_{nov}$ is the set of novel terms used to tag the texts in the novel term set. It was made up of alternative labels of AGROVOC concepts not contained in $\mathscr{L}_{syn}$ and whose preferred labels were not contained in $\mathscr{L}_{pref}$. If c_i is an AGROVOC concept with preferred label $pref_{c_i}$ and an alternative label alt_{c_i}, we have $\mathscr{L}_{syn} = \{alt_{c_i} | pref_{c_i} \in \mathscr{L}_{pref}\}$ and $\mathscr{L}_{nov} = \{alt_{c_i} | alt_{c_i} \notin \mathscr{L}_{syn} and pref_{c_i} \notin \mathscr{L}_{pref}\}$. Based on these definitions, we have $\mathscr{L}_{syn} \bigcap \mathscr{L}_{nov} = \emptyset$. For $\mathscr{L}_{syn}$ and $\mathscr{L}_{nov}$, we focused only on unigram synonym and novel terms to minimize the risk of overlaps with $\mathscr{L}_{pref}$. This decision does not mean that we extracted single words from multi-word terms to use them as agriculture-related terms themselves. It would lead to a loss of meaning and impact the prediction capability of the language models used for our term extraction task. We rather focused on single-word terms to avoid the occurrence of a term from $\mathscr{L}_{syn}$ or $\mathscr{L}_{nov}$ into $\mathscr{L}_{pref}$. To achieve this, we implemented automated checks to ensure that none of the unigram terms in $\mathscr{L}_{syn}$ or $\mathscr{L}_{nov}$ occurred in $\mathscr{L}_{pref}$ either as a term itself or as part of a multi-word term. For instance, if "precision agriculture" is an

[5] https://agrovoc.fao.org/sparql.

[6] The preferred labels of the AGROVOC concepts used to tag texts in the training, validation, and memorization term sets are listed in the "pref-labels-used.txt" file available in the "labels" folder in the repository.

agricultural term in $\mathscr{L}_{pref}$, "agriculture" does not appear either in $\mathscr{L}_{syn}$ or in $\mathscr{L}_{nov}$. If multi-word terms were to be considered for $\mathscr{L}_{syn}$ and $\mathscr{L}_{nov}$, we would have to ensure that: (i) the multi-word term does not appear in $\mathscr{L}_{pref}$; and (ii) none of the words in the term appear as terms themselves or parts of (other) multi-word terms in $\mathscr{L}_{pref}$. To better illustrate the second case, "precision farming" would be added as a term in $\mathscr{L}_{syn}$ or $\mathscr{L}_{nov}$ if: (i) "precision" did not appear as part of a term in $\mathscr{L}_{pref}$; and (ii) "farming" was not part of a multi-word term in $\mathscr{L}_{pref}$ or a term itself. Considering the complexity inherent to this case, and thus in the rules needed to perform automated overlap checks, we decided to not consider multi-word terms for synonym and novel term identification. Given the variety of lexical forms in which terms may occur in text, overlaps between $\mathscr{L}_{syn}/\mathscr{L}_{nov}$ and $\mathscr{L}_{pref}$ might still occur even after the application of automated overlap checks based on well-designed rules.

Based on the above, we have $\mathscr{L}_{syn} \cap \mathscr{L}_{pref} = \varnothing$ and $\mathscr{L}_{nov} \cap \mathscr{L}_{pref} = \varnothing$. The number of synonym terms used to tag the texts in the synonym term set was 178. The number of novel terms used to tag the texts in the novel term set was 461. The term sets we used to test synonym and novel term identification were relatively small. However, we avoided bias in measuring performance. The alternative labels of the AGROVOC concepts used as synonym and novel terms were retrieved by using a SPARQL query similar to the that used for the preferred labels. The SPARQL queries for the retrieval of preferred and alternative labels of AGROVOC concepts are available in the paper's GitHub repository. The lists of the alternative labels of AGROVOC contained in $\mathscr{L}_{syn}$ (used for the tagging of texts in the synonym term set) and $\mathscr{L}_{nov}$ (used for the tagging of texts in the novel term set) can also be found in the paper's GitHub repository.[7]

3.2. Text pre-processing and tagging

The text used for the training and evaluation of our system (text extracted from abstracts of publications indexed in AGRIS) was pre-processed to: (i) address quality issues; and (ii) enable text tagging. One of the main quality issues of the text was the existence of sequences of words munged together with no spaces in between them. For instance, a string in one of the texts was "partnershipshouldhavemembers", consisting of four words munged together into one word. To address this issue, a Python library doing probabilistic word splitting was used.[8] Words which were not acronyms were converted to lowercase. All the words (words found in the text as well as used as AGROVOC concept labels) were lemmatized. Lemmatization was considered to capture terms whose occurrence in text and AGROVOC labels might take different forms (e.g., the AGROVOC concept having "robots" as its preferred label can occur as "robot" in text). To lemmatize words, we used the Lemmatizer class[9] of the spaCy library (Honnibal and Montani, 2017). Lemmatizer assigns a base form to each word (the lemma of the word) using rules that are based on part-of-speech tags. Lemmatization is one of the components of spaCy's NLP pipeline (also containing part-of-speech tagging, dependency parsing, and NER components) and can be used either standalone or as part of the broader pipeline. To implement lemmatization, spaCy converts input text to a corpus of word vectors. In case of large-size texts, vectorization may have an increased cost in terms of the time needed. However, compared to other libraries providing lemmatization capabilities (e.g., the Python NLTK library's WordNet lemmatizer), spaCy's Lemmatizer class performs well also having the advantage of internally handling the part-of-speech tagging task which is a pre-requisite for lemmatization. Each AGRIS abstract was split into sentences. After pre-processing, the sentence's text was automatically tagged for the occurrence of agricultural terms in it using the preferred and alternative labels of AGROVOC concepts as described in Section 3.1. Tagging was made using the IOB scheme (Ramshaw and Marcus, 1999). The tag B-Agr was used to either indicate a single-word term, or the initial word in a multi-word term. I-Agr was the tag used to indicate a word being internal to a term. Finally, O was the tag used for denoting words not being terms at all.

To better illustrate how text pre-processing and tagging worked, let us assume that "Armillaria is a soil-borne fungus that causes a root and trunk rot of avocado" is one of the sentences in our training set. The version of the sentence obtained as the output of the lemmatization process is "armillaria be a soil-bear fungus that cause a root and trunk rot of avocado" (part of text pre-processing in our research was also the lower casing of words except for those being acronyms). Using the lemmas of the preferred labels of AGROVOC concepts, we identify "Armillaria", "soil-borne fungi" ("soil-borne fungi" is the concept preferred label available in AGROVOC, which is converted to "soil-bear fungus" after lemmatization), "trunk", and "avocado" ("trunks" and "avocados" are the preferred labels of AGROVOC concepts lemmatized to "trunk" and "avocado" respectively) as agriculture-related terms appearing in the sentence. Based on the above, "Armillaria", "trunk", and "avocado" are tagged as B-Agr. In the case of "soil-borne fungi", the word "soil-borne" is tagged as B-Agr and "fungi" as I-Agr. The other words in the sentence are tagged as O. The version of the sentence included in the training set was the original, not the lemmatized one. The flowchart shown in Fig. 1 provides a diagrammatic illustration of the pre-processing and tagging of texts extracted from AGRIS abstracts.

The tagged texts were converted to tuples $\left(\langle w_{i,1}, w_{i,2}, \cdots, w_{i,|s_i|}\rangle, \langle l_{i,1}, l_{i,2}, \cdots, l_{i,|s_i|}\rangle\right)$, where $w_{i,j}$ ($j = 1, 2, \cdots, |s_i|$) are the words making up a sentence s_i of length $|s_i|$ (number of words in the sentence) and $l_{i,j}$ ($j = 1, 2, \cdots, |s_i|$) is the IOB tag assigned to $w_{i,j}$. The text pre-processing and tagging work described in this section served the purpose of preparing our datasets (analytically presented in Section 4.1) for the fine-tuning of the BERT models. Details and statistics about the sentences that made up our training, validation, and test sets (min, max, and average sentence length), as well as the number of terms (i.e., agriculture-related entities tagged using labels of AGROVOC concepts) found in sentences (number of terms per sentence, as well as average number of unigram and multi-gram terms tagged per sentence) are provided in Section 4.1 ("Datasets"). More experiment-related information, such as the size of the batches used and a rationale for batch size selection, are provided in Section 4.2 ("Experimental setup").

3.3. Fine-tuning BERT for automatic term extraction

The automatic extraction of agriculture-related terms from text is treated as a sequence classification task where each word in a sentence is categorized into one of the {B-Agr, I-Agr, O} classes. By using BERT, term extraction takes the form of a token rather than word classification task. This is because BERT uses the WordPiece tokenizer, which tokenizes input based on a sub-word approach. This means that OOV words are split into more than one tokens if they are not contained in the model's vocabulary (Schuster and Nakajima, 2012). In the case of a word being split into multiple sub-words, the IOB label indicating whether the word is (part of) an agricultural term, or not, will be eventually given to the first of the sub-words that make up the word (Devlin et al., 2018). This approach does not change the total number of labels in each of the classes of the classification task, and thus does not impact the model's performance in ATE (as it would be the case from the introduction of additional IOB labels if label assignment was taking place for each of the sub-words). For instance, the term "deforestation" will be tokenized to "def", "##ores", and "##tation". The label B-Agr

[7] The alternative labels of the AGROVOC concepts used to tag the texts in the synonym term set ($\mathscr{L}_{syn}$) are listed in the "synonym-labels-used.txt" file available in the "labels" folder in the repository. The alternative labels of the AGROVOC concepts used to tag the texts in the novel term set ($\mathscr{L}_{nov}$) are listed in the "novel-labels-used.txt" file available in the "labels" folder in the repository.

[8] https://github.com/keredson/wordninja.

[9] https://spacy.io/api/lemmatizer.

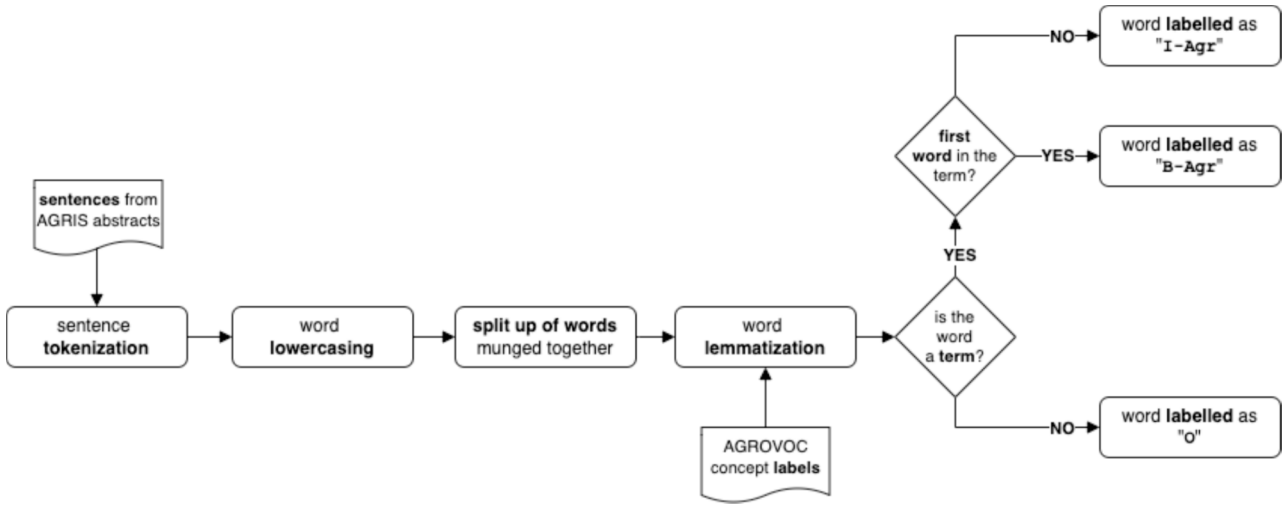

Fig. 1. Pre-processing and tagging of the texts in the training, validation, and test sets using the labels of AGROVOC concepts and the IOB scheme.

assigned to it as part of our corpus' dataset creation will be automatically given to the sub-word "def" (the first of the sub-words in which the term was split). The other sub-words will be assigned a dummy label ("-100").

Fig. 2 illustrates our ATE's architecture which consists of three layers: (i) the embedding layer; (ii) the stack of encoder layers; and (iii) the classification layer.

The embedding layer outputs an embedding for each token in the sentence as the sum of the embedding of the tokens generated by the tokenizer, the embedding encoding the token's position in the sentence, and the embedding encoding the sentence the token belongs to. In the case of $BERT_{BASE}$ (size of all the models compared in our research), each token embedding has a dimension of 768 with each dimension encoding a latent semantic feature of the token (Devlin et al., 2018). The dimension of the vector being input to each encoder layer, as well as outputted by it, is also 768 (Devlin et al., 2018). The embedding and encoder layers are part of the pre-trained language model. The classification layer added on top of the pre-trained language model layers is responsible for the classification of each token into one of the considered classes and therefore the identification of a word in the text as (part of) an agriculture-related term. It is implemented using the BertForTokenClassification[10] class, which is a sub-class of the torch.nn.Module PyTorch class. The classification layer receives a 768-dimensional vector as input and outputs a 3-dimensional vector of the token's probability to belong to each of the considered classes. In this paper, we compare Agriculture-BERT to Sci-BERT, RoBERTa, and vanilla BERT. RoBERTa and vanilla BERT were considered as baselines to obtain insights into whether and how much better a language model pre-trained on specialized text corpora performs when used for automatic agricultural term extraction.

3.3.1. Sci-BERT

Sci-BERT is a language model pre-trained on scientific literature using vanilla BERT as a checkpoint for further pre-training (Beltagy et al., 2019). The corpus used for its pre-training is made up of 1.14 M research papers available in Semantic Scholar, which are about computer science (18 % of the corpus) and biomedicine (82 % of the corpus). The total number of tokens in the corpus is 3.17 billion. Cased and uncased versions of Sci-BERT are available in the Hugging Face library.[11]

3.3.2. Agriculture-BERT

Agriculture-BERT is a BERT model pre-trained on agricultural texts, using Sci-BERT as a checkpoint. Agriculture-BERT comes shipped with the Hugging Face library.[12] Paragraphs taken from agricultural texts (e.g., texts available in the National Agricultural Library of the US Department of Agriculture[13]) form the corpus on which the model was pre-trained based on the self-supervised Masked Language Modeling task (Quadros, 2021). Farming, plant diseases, pest control, food and nutrition, and fertilizers are some of the topics addressed in the corpus texts.

3.3.3. RoBERTa

Starting from the observation that the pre-training process of BERT results in model undertraining, Liu et al. (2019) propose RoBERTa; a version of vanilla BERT pre-trained by: (i) using more textual data and larger text batches; (ii) ignoring the next sentence prediction objective; (iii) using longer sequences of tokens; and (iv) dynamically changing the token masking pattern applied to training data. A ready-to-deploy implementation of RoBERTa can be found in Hugging Face.[14]

4. Experiments and results

4.1. Datasets

To the best of our knowledge, there are no benchmark datasets for evaluating performance in automatic agricultural term extraction in English. Hence, it was decided to create and use a silver standard corpus. The silver standard corpus approach adopted in our research is different to the approaches followed in previous research, where gold standard corpora are used (not being, however, well-known benchmarks or available for re-use). Our silver standard corpus $\mathscr{C}$ consisted of a training set, a validation set, and three test sets. It was created using the texts extracted from publication abstracts available in the Food and Agriculture Organization's AGRIS database. AGRIS does not deliver full publication texts, but rather serves as an index providing access to abstracts of the indexed publications. If a full text version of an indexed

[10] https://huggingface.co/transformers/v3.0.2/model_doc/bert.html#bertfortokenclassification

[11] https://huggingface.co/allenai/scibert_scivocab_cased and https://huggingface.co/allenai/scibert_scivocab_uncased respectively.

[12] https://huggingface.co/recobo/agriculture-bert-uncased.

[13] https://www.nal.usda.gov/.

[14] https://huggingface.co/roberta-base.

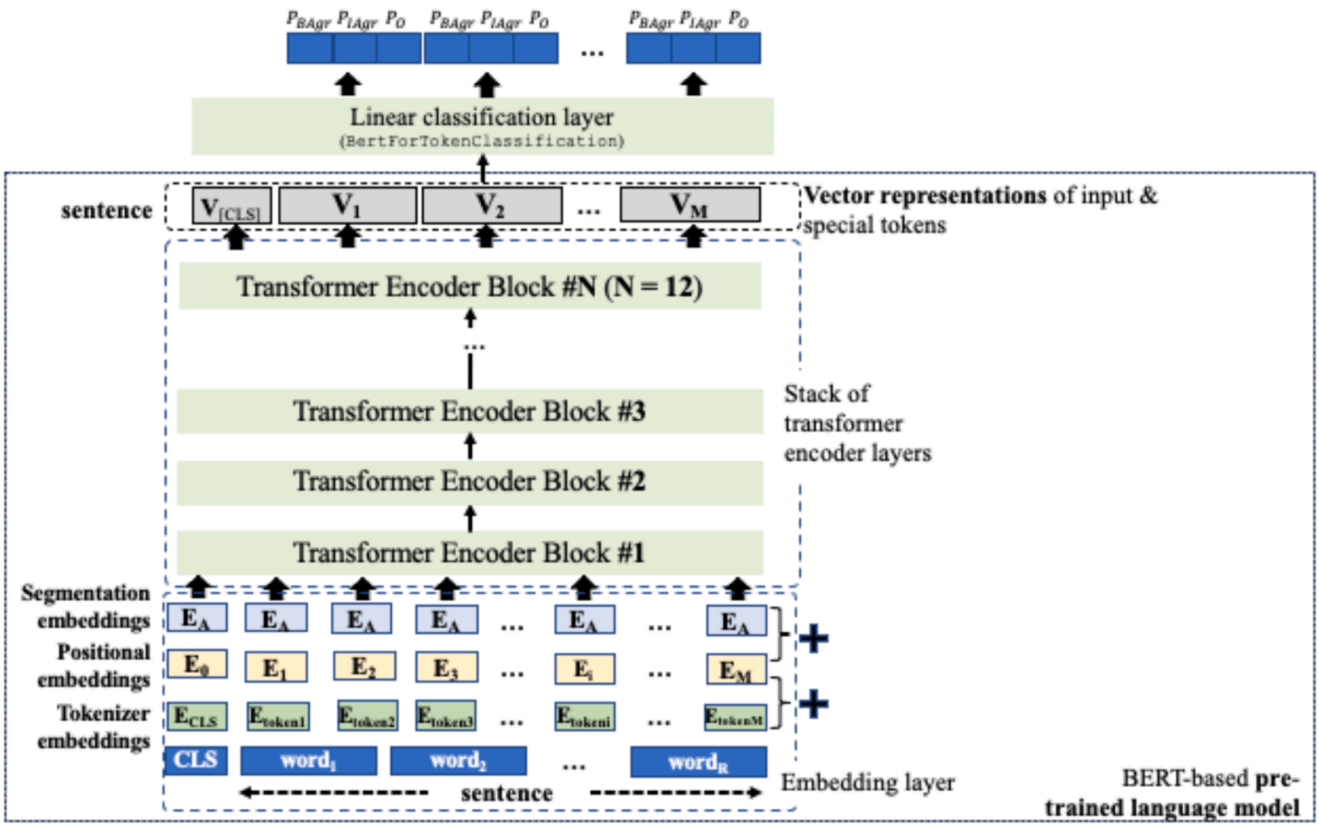

Fig. 2. Our BERT-based ATE architecture.

publication is available in another repository, a link is provided. AGRIS publications include books, journal and conference papers, monographs, datasets, and grey literature (e.g., theses, dissertations, unpublished scientific and technical reports, and book chapters) related to agriculture and food.

Following the approach presented by Kim and Kang (2022), we used three test sets. Test set 1 was the memorization term set containing texts that were tagged using terms also seen during training (a subset of the preferred labels of AGROVOC concepts in the $\mathscr{L}_{pref}$ set described in Section 3.1). Test set 2 was the synonym term set, which consisted of texts tagged with terms being synonyms of those seen during training (the terms contained in the $\mathscr{L}_{syn}$ set described in Section 3.1). Test set 3 was the novel term set comprising texts that were tagged with terms neither seen during training nor being synonyms of those seen during training (the novel terms contained in the $\mathscr{L}_{nov}$ set described in Section 3.1). Considering that agriculture-related terms have synonyms (according to AGROVOC "precision agriculture" and "precision farming" are synonym terms), as well as that agricultural terminology evolves over time, we used test sets 2 and 3 to obtain insights into our ATE's generalization capacity.

For our training, validation, and memorization test sets, we used approximately 14.5 K sentences from abstracts randomly selected from the AGRIS database. Our training, validation, and memorization set splits were created following the 80–10-10 rule. The training set was made up of approximately 11.5 K sentences and the validation and memorization test sets of almost 1.5 K sentences each. For the synonym and novel term sets, we considered two additional hold-out sets of 1.2 K sentences each, other than those in the training, validation, and memorization sets. By using $\mathscr{L}_{syn}$ and $\mathscr{L}_{nov}$ as sets of agricultural terms to tag the texts in the synonym and novel term sets respectively ($\mathscr{L}_{syn} \cap \mathscr{L}_{pref} = \varnothing$ and $\mathscr{L}_{nov} \cap \mathscr{L}_{pref} = \varnothing$), we ensured no term overlap between those two sets and the training set. A summary of our datasets and the sets of agriculture-related terms used to tag them is shown in

Table 2
Datasets constituting our silver standard corpus.

Dataset	Number of sentences	Terms used to tag text	Dataset	Number of sentences	Terms used to tag text
Training set	11.5 K	$\mathscr{L}_{pref}$	Synonym term set	1.2 K	$\mathscr{L}_{syn}$
Validation set	1.5 K	$\mathscr{L}_{pref}$	Novel term set	1.2 K	$\mathscr{L}_{nov}$
Memorization set	1.5 K	$\mathscr{L}_{pref}$			

Table 2. Table 3 presents features of the sentences in our silver standard corpus. All datasets are available in the GitHub repository of the paper.

The construction pipeline of our silver standard corpus is illustrated in Fig. 3.

Table 3
Features of the sentences in our silver standard corpus and their values.

Sentence feature	Value	Sentence feature	Value
Max sentence length (in words)	85	Max number of terms in a sentence	19
Min sentence length (in words)	5	Min number of terms in a sentence	1
Avg sentence length (in words)	18.3	Avg number of terms in a sentence	2.7
		Avg number of unigram terms in a sentence	2.4
		Avg number of multi-gram terms in a sentence	0.3

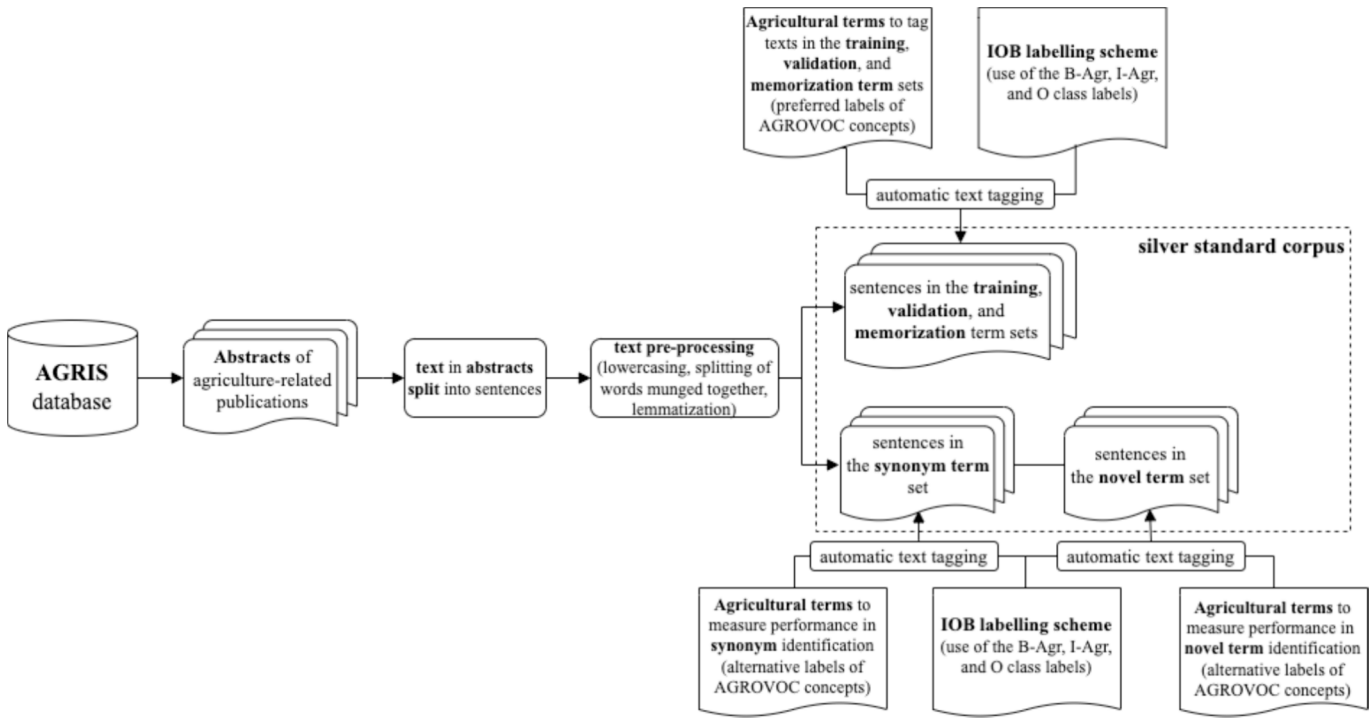

Fig. 3. Silver standard corpus construction pipeline.

4.2. Experimental setup

The architecture of all the compared language models was based on the $BERT_{BASE}$ size configuration: L (number of stacked encoder layers) = 12, H (size of hidden layers) = 768, A (number of self-attention heads) = 12, and P (number of trainable parameters) = 110 M (Devlin et al., 2018). In our research, five experiments were performed. In each experiment, a different split of the 14.5 K sentences in our training, validation, and memorization term sets was created randomly. Performance in the identification of terms also seen during training was measured using the memorization term set created in each of the experiments. Performance in synonym and novel term identification was measured once based on the synonym and novel term sets. To compare the BERT models considered in our study, we used the same configuration scenario regarding the model layers kept frozen, or being updated, during training. That scenario involved keeping the embedding layer frozen and updating the weights of all the encoder layers (hereinafter, basic configuration scenario). We made this decision because the contextual information encoded in the embeddings when BERT is pre-trained on large text corpora is rich (Koren, 2021). Thus, no major changes in context capturing will probably occur when updating the embedding layer during BERT's fine-tuning on a specific task (especially if fine-tuning is made using a corpus that is relatively small compared to the corpus used for the model's pre-training).

For the best-performing model among those compared, we expanded our experiments to investigate the effect of different fine-tuning configuration scenarios on the agricultural term extraction performance. According to Merchant et al. (2020), the performance achieved in the case of fine-tuning BERT on the tasks of Natural Language Understanding and Question Answering does not drop significantly when updating the top 4 encoder layers, compared to having all encoder layers updated. Such results indicate that the top encoder layers are those learning task-specific patterns, as opposed to the bottom layers which appear to learn more generic, task-agnostic patterns. The effect of keeping the embedding layer frozen, or updating it, during fine-tuning has been less researched. Some evidence is provided by Koren (2021) according to which small performance improvements occur when keeping the embedding layer frozen, particularly in large vocabulary cases. The fine-tuning configuration scenarios considered in our research are presented in Table 4.

The hyperparameter configurations used in our experiments and a brief rationale are presented in Table 5.

All experiments were executed on a workstation using a GeForce RTX 3090 with CUDA GPU, which provides a better processing speed for the development of models based on using transformers. A Core AMD Ryzen Threadripper 1950x CPU with a 16-core processor was used for the machine learning training. The workstation runs on the Ubuntu 18.04 operating system.

4.3. Performance evaluation

4.3.1. Comparison of the selected BERT models on the memorization term test set

The metrics used to evaluate the performance of the models compared in our research are precision (P), recall (R), and F1-score (F1). Performance was measured both at the term and class levels. Considering the imbalanced nature of our training and evaluation datasets (the instances of the O class are by 65.34 % more than the instances of the B-Agr and I-Agr classes), the macro-averaged F1-score was used. Class

Table 4
Fine-tuning configuration scenarios investigated for their impact on agricultural term extraction performance for the best-performing BERT-based model.

		stack of encoder layers	
		all encoder layers updated	top 4 encoder layers updated
embedding layer	embedding layer updated	embedding layer updated + all encoder layers updated	embedding layer updated + top 4 encoder layers updated
	embedding layer frozen	embedding layer frozen + all encoder layers updated	embedding layer frozen + top 4 encoder layers updated

Table 5
Hyperparameter configurations and rationale.

Hyperparameter and value assigned	Rationale
Optimization algorithm = SGD	ADAM was also tested with different learning rate values, but the performance was consistently poor.
Learning rate = 1e-3	The learning rates used by Devlin et al. (2018) for the fine-tuning of BERT on different machine learning tasks ranged from 2e to 5 to 5e-5. Smaller values of learning rate were also tested in our experiments without leading to improvements in performance. Higher learning rate values did not allow the model to converge.
Learning rate reduction factor = 0.8 (ReduceLROnPlateau)	Reduction of the learning rate value by a factor of 0.8 after 5 epochs if the model is not learning. The value suggested in literature is 0.1; in our research, this value was found to randomly change the training behavior.
Dropout rate = 0.35	This is the highest dropout rate found to prevent overfitting; higher dropout rate values did not prevent overfitting.
Batch size = 1	The batch size used by Devlin et al. (2018) is 32. In our experiments, it was observed that the model was learning slowly when higher batch sizes were tested. In addition, the model performance was poor.
Patience = 10	The number of epochs to wait after the model stops improving its performance on the validation set. The patience value in our research was empirically selected.

imbalance is a common characteristic of datasets used in NER and ATE (Li et al., 2019). In such a case, macro-averaged F1 is more appropriate given that it treats all classes equally as opposed to the weighted F1-score (per class contribution to the F1-score average is weighed by the size of the class) and micro-averaged F1-score (it considers the TPs, FNs, and FPs of all classes) (Leung, 2022). Table 6 shows each model's performance, achieved on average in each of our experiments, using the memorization term set split that was randomly created in each of the five experiments.

Agriculture-BERT performs better than the rest of the BERT models and Sci-BERT is the second-best performing model. Agriculture-BERT's precision is higher than the other models' precision from 1.39 % to 1.66 %. The recall of Agriculture-BERT is higher than that of the other models in a range from 1.39 % (difference to Sci-BERT's recall) to 3.36 % (difference to vanilla BERT's recall). Differences between Agriculture-BERT's F1-score and the F1-scores of the other BERT models range from 1.48 % to 2.99 %. Differences in performance between Agriculture-BERT and Sci-BERT (best and second-best models in terms of performance), in all metrics, are more than 1.3 %. When comparing Sci-BERT to RoBERTa, as well as RoBERTa to vanilla BERT, performance differences are not higher than 1.05 % in any metric. Table 7 shows the performance of the models compared in our research regarding their ability to identify raw instances of the B-Agr, I-Agr, and O classes in text.

Agriculture-BERT has better performance compared to the other models except for the recall of the B-Agr class, where RoBERTa performed better. RoBERTa's F1-score for the B-Agr class is also higher than the F1-score of the B-Agr class of Agriculture-BERT. RoBERTa also exhibits better performance regarding the precision of the O class compared to Agriculture-BERT, yet at a marginal level (0.12 %). The results in Table 7 also show that there are big differences, for all models, between: (i) the precision of B-Agr and I-Agr instance identification; (ii) the recall of B-Agr and I-Agr instance identification; and (iii) precision and recall of I-Agr instance identification. These differences provide indications for errors in agriculture-related term identification and cases in which our BERT-based ATE did not work well. A detailed explanation of those results is provided in Section 5 ("Discussion").

Table 6
Prediction performance achieved at the term level on average by each of the compared BERT models.

Language model	Precision (P)	Recall (R)	F1 score (F1)
Agriculture-BERT	**85.28**	**77.22**	**80.60**
Sci-BERT	83.89	75.83	79.12
RoBERTa	83.66	75.06	78.07
Vanilla BERT	83.62	73.86	77.61

(*) Performance is reported as % rate.

4.3.2. Comparison of the selected BERT models on the synonym and novel term test sets

Performance in synonym and novel term identification was evaluated using recall. Table 8 presents the results from the evaluation of all BERT models.

Agriculture-BERT outperforms all the other models in novel term identification with the differences in performance being at least 8.00 % (difference from vanilla BERT, the second best-performing model). In synonym term identification, Sci-BERT had the best performance. However, the difference from the performance of Agriculture-BERT is marginal (0.49 %). Hence, we can claim that Agriculture-BERT is almost as good as Sci-BERT in synonym term identification.

4.3.3. Evaluation of performance in different fine-tuning scenarios

Agriculture-BERT had the best performance in the identification of terms also seen during training and novel terms. Regarding synonym term extraction, Agriculture-BERT was almost as good as Sci-BERT. Given these results, performance evaluation based on the fine-tuning configuration scenarios of Table 4 was made for Agriculture-BERT. The results that were obtained for the various scenarios when using the memorization term set are shown in Table 9. The underlined figures indicate the performance related to the basic configuration scenario (embedding layer frozen + all encoder layers updated). The figures in bold indicate the best performance per evaluation metric.

As shown in Table 9, the update of the embedding layer and all the encoder layers of Agriculture-BERT resulted in better performance compared to that achieved in the case of the basic configuration scenario. Precision improved by 4.49 % and recall by 6.04 %. These improvements led to a 5.54 % increase in F1-score. In the scenario "embedding layer updated + top 4 encoder layers updated", performance did not actually improve compared to the basic configuration scenario. There was only a marginal improvement in precision by 0.31 %. The scenario "embedding layer frozen + top 4 encoder layers updated" resulted in a decrease in performance compared to the basic configuration scenario. The results of performance evaluation at the class level are shown in Table 10.

As expected, the performance achieved in the case of the "embedding layer updated + all encoder layers updated" configuration scenario was better than that of the basic configuration scenario. Differences in performance in the identification of B-Agr and I-Agr class instances remain with the precision, recall, and F1-score of the B-Agr class being higher than those of the I-Agr class. However, the differences observed are lower than those in the case of the basic configuration scenario. The gap between precision and recall in I-Agr class instance identification was also reduced.

Table 11 shows the performance for each of the fine-tuning configuration scenarios regarding synonym and novel term identification. Recall is the evaluation metric used.

When measuring performance in synonym and novel term identification, the "embedding layer updated + all encoder layers updated" configuration scenario was not the best performing one. Best performance in synonym term identification was achieved in the case of the basic configuration scenario. For novel term identification, the best performance was obtained for the scenario "embedding layer updated + top 4 encoder layers updated". However, differences with the configuration scenarios having the second and third best performance were marginal (0.05 % with "embedding layer updated + all encoder layers

Table 7
Per class prediction performance achieved on average in the conducted experiments by each of the BERT models.

Language model	Precision (P)			Recall (R)			F1-score (F1)		
	B-Agr	I-Agr	O	B-Agr	I-Agr	O	B-Agr	I-Agr	O
Agriculture-BERT	**87.93**	**71.40**	96.50	83.30	**50.35**	**98.02**	85.54	**59.00**	**97.25**
Sci-BERT	87.02	68.33	96.33	83.00	46.70	97.78	84.95	55.37	97.04
RoBERTa	87.24	67.12	**96.62**	**85.60**	41.91	97.67	**86.08**	51.11	97.02
Vanilla BERT	87.20	67.88	95.77	82.59	41.38	97.59	84.81	51.34	96.67

(*) Performance is reported as % rate.

Table 8
Performance of the compared BERT models in the identification of synonym and novel terms.

Language model	Synonym term identification	Novel term identification
Agriculture-BERT	77.98	**59.51**
Sci-BERT	**78.47**	50.69
RoBERTa	74.58	48.36
Vanilla BERT	71.32	51.52

(*) Performance is reported as % rate.

Table 9
Performance achieved by agriculture-BERT on average at the term level in each of the fine-tuning configuration scenarios (memorization term set).

Fine-tuning configuration scenario	Precision (P)	Recall (R)	F1 score (F1)
embedding layer updated + all encoder layers updated	**89.77**	**83.26**	**86.14**
embedding layer updated + top 4 encoder layers updated	85.59	76.76	80.51
embedding layer frozen + all encoder layers updated (basic configuration scenario)	85.28	77.22	80.60
embedding layer frozen + top 4 encoder layers updated	81.61	72.56	76.37

(*) Performance is reported as % rate.

updated" and 0.61 % with the basic configuration scenario).

5. Discussion

The evaluation results obtained for Agriculture-BERT show that it performs better, or almost as good as the other BERT models compared in our research, when fine-tuned on the ATE task. Comparing the performance of Agriculture-BERT, in all our test datasets, to the performance of the rule-based ATE methods and tools listed in Table 1, we observe better performance scores for Agriculture-BERT. The precision and recall of the rule-based ATE studies cited in Section 2 are lower than Agriculture-BERT's average performance (using the basic configuration scenario) on the memorization term set (P = 85.28 % and R = 77.22 %). Similar conclusions are drawn for Agriculture-BERT's performance in synonym and novel term identification. Recall in those cases was 77.98 % and 59.51 % respectively. These scores are higher than the recall of the rule-based methods reviewed in our research. Likewise, the results obtained when using the various fine-tuning configuration scenarios in Table 4 (apart from the basic configuration scenario) reveal that the performance of Agriculture-BERT is better, or at least comparable, to that of rule-based ATE. When comparing the performance of Agriculture-BERT, using any of the configuration scenarios, to that of the deep learning approaches cited in our research, we conclude that Agriculture-BERT does better than the methods proposed by Manjunath and McCrae (2021) and Panoutsopoulos et al. (2022). The performance of Agriculture-BERT for the task of identifying terms also seen during training using the "embedding layer updated + all encoder layers updated" fine-tuning configuration scenario is comparable to the performance reported by Espejo-Garcia et al. (2019). No direct comparisons can be made to the work of Guo et al. (2022) given its different linguistic context. Based on the above, we can consider the results reported in our research and the performance of Agriculture-BERT in agriculture-related ATE as acceptable.

The results shown in Table 7 reveal big differences in the performance of the compared BERT models regarding the identification of instances of the B-Agr class and the I-Agr class. These differences are indicative of cases in which the BERT models were not successful in correctly identifying agriculture-related terms in text. The performance of all models in the identification of instances of the B-Agr class was higher than the performance in the identification of I-Agr class instances. This means that there was a bias towards the identification of unigram terms, which had an impact on the prediction capacity of our ATE favoring the identification of unigram terms. In other words, in cases in which a word being a term itself occurs in the text as part of a multi-gram term, BERT will tend to predict the unigram rather than the multi-gram term. For instance, the word "farming" is a term by itself

Table 11
Performance for the identification of terms not seen during training.

Fine-tuning scenario	Synonym term identification	Novel term identification
embedding layer updated + all encoder layers updated	72.94	60.07
embedding layer updated + top 4 encoder layers updated	69.25	**60.12**
embedding layer frozen + all encoder layers updated (basic configuration scenario)	**77.98**	59.51
embedding layer frozen + top 4 encoder layers updated	62.87	52.87

(*) Performance is reported as % rate.

Table 10
Per class performance achieved by agriculture-BERT on average in each of the fine-tuning configuration scenarios.

Fine-tuning configuration scenario	Precision (P)			Recall (R)			F1-score (F1)		
	B-Agr	I-Agr	O	B-Agr	I-Agr	O	B-Agr	I-Agr	O
embedding layer updated + all encoder layers updated	**91.70**	**80.17**	**97.45**	**88.16**	**62.97**	**98.64**	**89.89**	**70.48**	**98.04**
embedding layer updated + top 4 encoder layers updated	89.18	71.69	95.89	80.78	51.16	98.32	84.77	59.68	97.09
embedding layer frozen + all encoder layers updated (basic configuration scenario)	87.93	71.40	96.50	83.30	50.35	98.02	85.54	59.00	97.25
embedding layer frozen + top 4 encoder layers updated	84.91	65.39	94.52	73.88	46.20	97.59	79.01	54.05	96.03

(*) Performance is reported as % rate.

(alternative label of the concept uniquely identified in AGROVOC by the URI http://aims.fao.org/aos/agrovoc/c_2807) but also occurs as part of the term "precision farming" (alternative label of the concept in AGROVOC with the URI: http://aims.fao.org/aos/agrovoc/c_92363). Considering the bias towards the identification of unigram terms, in the above example, the BERT models will most probably return "farming" instead of "precision farming", which is the term that should have been identified. This bias, and hence error in term identification, can be attributed to the imbalance in the number of unigram and multi-gram terms in our datasets (unigrams are 8 times more than multi-grams).

Another case in which our BERT-based ATE was not successful is indicated by the differences between the precision and recall of I-Agr instance identification. These differences show a misidentification of the boundaries of multi-gram terms. In other words, one (or more) of the words in a multi-word term may be missed and not predicted by our BERT-based term extractor. Based on our experiments' results, the number of words falsely identified as not being instances of I-Agr (FNs) is higher than the number of words falsely classified as I-Agr class instances (FPs). This means poorer performance in the correct identification of actual instances of the I-Agr class compared to how BERT distinguishes actual from false I-Agr instances. As a result, we get errors in the identification of multi-word terms. The higher number of unigram terms in our datasets, compared to multi-gram terms, can explain the occurrence of such errors.

When testing all the BERT models considered in our research in synonym and novel term identification, performance drops. Similar results were obtained for Agriculture-BERT when different configuration scenarios were tested. Reduction in synonym and novel term identification performance is expected and is also supported by the findings of other research studies (Kim and Kang, 2022). The decrease in recall means that BERT's capacity in correctly identifying synonym and novel agricultural terms in text drops. Performance drops can be attributed to the fact that the context within which synonym and novel terms occur in text differs from the context in which terms in the training set appear. As a result, BERT failed in correctly identifying words being agriculture-related terms (decrease of the number of TPs and/or increase of the number of FNs). However, we need to consider the small size of the synonym and novel term sets (1.2 K sentences in each set) as well as the small number of unigram terms used to label them. More research, not just based on unigram terms, is needed for further insights into BERT's capacity to identify terms not seen during training.

For the impact of the fine-tuning configuration scenarios on term extraction performance, we found a different scenario being the best performing one for each of the term extraction cases considered in our research (identification of terms also seen during training, synonym terms, and novel terms). In the case of measuring performance on the memorization term test set, the per class prediction capacity of Agriculture-BERT improved across all evaluation metrics when updating the embedding and all the encoder layers during training. This means that Agriculture-BERT becomes better in identifying single-word terms when all its layers (embedding layer and all the encoding layers) are updated during training compared to only updating the encoder layers (basic configuration scenario). It also becomes better in multi-word term identification. The increase in the precision and recall of the I-Agr class, compared to that of B-Agr, indicates that Agriculture-BERT's improvement in multi-word term identification is higher than its improvement in single-word term identification.

Regarding novel term extraction from text, the scenario "embedding layer updated + top 4 encoder layers updated" had the best performance. However, the difference from the scenario "embedding layer updated + all encoder layers updated" was marginal (0.05 %). This difference allows to claim that the scenario "embedding layer updated + all encoder layers updated" is equally good to the "embedding layer updated + top 4 encoder layers updated" scenario. We assume that the linguistic patterns learned when updating all the encoder layers, as opposed to updating only the top 4 layers, does not make a difference. However, having just the top 4 layers updated is less resource demanding given the smaller number of parameters being updated. As a result, the computational cost is lower (85 M parameters need to be learned when all the encoder layers become updated as opposed to 28 M parameters in the case of updating the top 4 encoder layers).[15]

The basic configuration scenario had the best performance in synonym term identification. In addition, the difference of the basic configuration scenario's performance in novel term identification compared to that of the best performing configurations ("embedding layer updated + top 4 encoder layers updated" and "embedding layer updated + all encoder layers updated") was less than 1 % (59.51 % compared to 60.12 % and 60.07 % respectively). The contextual information encoded and learned from the update of the embedding layer does not lead to a big improvement in the capacity of Agriculture-BERT to extract synonym and novel terms. Based on these results, we may conclude that the basic configuration scenario makes a good candidate for synonym and novel term identification. Keeping the embedding layer frozen allows to save resources that would be otherwise needed for learning the embedding layer's 24 M parameters.[16] Further research, also based on the use of multi-gram terms for synonym and novel term, will allow us to draw more sound conclusions on the impact of the various configuration scenarios on BERT's term extraction performance.

6. Conclusions and future work

Research on agriculture-related ATE has been limited. In our research, we compared the performance of Agriculture-BERT to that of Sci-BERT and two general-purpose BERT models (namely, RoBERTa and vanilla BERT) when fine-tuned on the automatic extraction of agricultural terms from English texts. We have shown that with a few exceptions, Agriculture-BERT performs better than the other models. We constructed a silver standard corpus (available in the GitHub repository of our paper) consisting of text datasets automatically tagged for the occurrence of agricultural terms in them, which was used to train and evaluate the compared models. Using AGROVOC as a standard for agricultural terminology allowed us to create our corpus for training and evaluation by eliminating inter-annotator disagreement in the text tagging task.

We have also used different fine-tuning configuration scenarios related to the BERT layers kept frozen, or being updated, during training to investigate their impact on term extraction performance. Our results show that different fine-tuning configurations achieve the best performance in each of our testing cases (performance in the identification of: terms also seen during training using the memorization term set; synonym terms using the synonym term set; and novel terms using the novel term set). When measuring performance in the identification of terms also seen during training, the configuration scenario in which all layers (embedding layer and all encoder layers) are updated is the best performing one. This means that the context learned when updating the embedding layer and all encoder layers has an effect on the identification of terms also seen during training. On the other hand, updating all the model layers during training appears to not have an impact on the identification of synonym and novel terms. To reach more rigorous conclusions, further research is needed extending the evaluation of Agriculture-BERT in multi-gram synonym and novel term extraction.

The use of texts only from AGRIS publication abstracts to train and evaluate the compared models was a limitation of our research. AGRIS delivers abstracts of publications related to agrifood, however more

[15] An explanation of the number of parameters learned at each encoder layer is provided at https://stackoverflow. com/questions/64485777/how-is-the-number-of-parameters-be-calculated-in-bert-model.

[16] A calculation of the number of parameters learned in the embedding layer is provided at https://stackoverflow. com/questions/64485777/how-is-the-number-of-parameters-be-calculated-in-bert-model.

sources of agricultural texts need to be considered in the future to build a corpus encompassing a broader range of agricultural content. The length of AGRIS abstracts is small and they have specific linguistic characteristics given that they are outputs of scientific work. Using texts from non-scientific sources as well will allow for the capture and encoding of a broader context of agricultural term occurrence in text. Another limitation was the construction of the silver standard corpus used in our research by just using AGROVOC as an agriculture-related terminology standard. This decision was limiting given that there are other vocabularies and ontologies (e.g., FoodOn) that can provide a broader coverage of the agrifood domain. For instance, "preservation by treatment with chemicals" is a concept in FoodOn having the URI: http://purl.obolibrary.org/obo/FOODON_03470108, not defined in AGROVOC. The imbalance in the number of single- and multi-word terms in our datasets was also one of the limitations of our research. It led to a bias in the ATE capacity of BERT favoring the prediction of unigram over multi-gram terms. This limitation can be attributed to using texts only from AGRIS, as well as the use of AGROVOC as the only source of agriculture-related terminology. The construction of a corpus from texts from a wide range of sources together with the tagging of the texts using terms defined in various domain vocabularies and ontologies has the potential to reduce imbalance in the occurrence of unigram and multi-gram term.

Improving ATE performance is a target for further research. To this end, future work will focus on the integration of knowledge captured in existing domain knowledge representations into our BERT-based agricultural term extraction with the aim to enhance the term extraction performance. In addition, we will consider the systematic comparison of ATE based on fine-tuning BERT models to LLMs with the aim to investigate whether the use of LLMs for ATE can lead to the achievement of significantly better performance. In our future work, we will also consider the use of texts from a range of sources. Sources to be considered are Wikipedia, the National Agricultural Library of USDA (i.e., the U.S. Department of Agriculture), as well as texts produced from the work in research and innovation projects related to agrifood. Following our silver standard corpus approach, a broader set of agrifood-related vocabularies and ontologies is intended to be considered for text tagging and the generation of the datasets for model training and evaluation.

CRediT authorship contribution statement

Hercules Panoutsopoulos: Writing – review & editing, Writing – original draft, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Borja Espejo-Garcia:** Writing – review & editing, Validation, Software, Resources, Investigation, Data curation. **Stephan Raaijmakers:** Writing – review & editing, Supervision. **Xu Wang:** Writing – review & editing, Methodology. **Spyros Fountas:** Writing – review & editing, Funding acquisition. **Christopher Brewster:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used in the paper and the code that was developed are available through the paper's GitHub repository.

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